

Sanjay Kumar Samala

✉ sanjaykumarsamala2001@gmail.com | 📞 (571)603-5359 | 🌐 sanju.samala | 📄 Sanjay kumar samala | 🔄 SanjuCyb3r

EDUCATION

Master of Science in Cyber Security Jan 2024 – Present
George Mason University, Fairfax, VA CGPA: 3.44
Relevant Coursework: Networks and Cyber Security, Penetration Testing, Malware Reverse Engineering, Cloud Computing Security, Mobile Device Forensics

Bachelor of Engineering in Electrical and Electronics Aug 2019 – Jul 2023
Osmania University, Hyderabad, India CGPA: 3.55

SKILLS

Offensive Security: Nmap, Hydra, Burp Suite, Metasploit, Nessus, Exploit Development, Vulnerability Assessment, Web Exploitation, Binary Exploitation, Red Teaming

Defensive Security: SIEM (Splunk, ELK), Zeek, Wireshark, YARA, SOC Monitoring, Incident Response, Risk Mitigation, Secure Coding, Threat Modeling, MITRE ATT&CK, NIST CSF, Zero Trust, CVSS Scoring, OWASP Top 10

Forensics & Malware: FTK Imager, Autopsy, IDA Pro, Hex Editor, Process Monitor, Process Explorer, FakeDNS, Registry Analysis, File Carving, IOC Extraction, C2 Detection, Reverse Engineering, Cryptography, Forensic Chain of Custody

Programming & Automation: Python, C, Powershell, Security Automation, Custom Tooling

EXPERIENCE

Graduate Teaching Assistant Aug 2024 – Present
George Mason University Fairfax, VA

- Leveraging the OSI reference model and advanced data link protocols, orchestrated interactive demonstrations of packet-switching and congestion control, resulting in improved conceptual retention across **250+** cybersecurity students.
- Utilizing **Burp Suite**, **Kali Linux**, and **Metasploit**, directed offensive security simulations and facilitated exploit development labs, enhancing technical competence and adversarial thinking by 20%.
- Applying **SSL/TLS** configuration principles with Zeek log analysis and **Splunk**, engineered secure communication environments for 5+ web applications, reducing exploitable vulnerabilities in encryption practices.
- Integrating **NIST Cybersecurity Framework** with human-computer interaction design, mentored students on **Zero Trust** methodologies and risk-aware decision-making, producing measurable improvements in secure coding and threat mitigation.

Penetration Testing Intern Sep 2023 – Oct 2023
HackersForYou Hyderabad, India

- Using adversarial testing methodologies and tools such as Nmap, Burp Suite, and Metasploit, executed penetration tests on **7+** enterprise **web applications**, resulting in the discovery and remediation of **30+ critical vulnerabilities** including privilege escalation and buffer overflow exploits.
- Applying **Python-based automation** for reconnaissance and vulnerability validation, engineered scripts that **reduced** manual testing effort by **25%**, enabling faster triage of high-risk findings and improving assessment efficiency.
- Leveraging **exploit development** techniques and post-exploitation frameworks, validated security flaws through controlled exploitation, which strengthened incident response readiness and **reduced** client-reported vulnerabilities by **40%**.
- Delivering technical reports with CVSS scoring and remediation roadmaps, collaborated with cross-functional stakeholders to align findings with security frameworks (**OWASP Top 10**, **MITRE ATT&CK**), resulting in measurable hardening of client systems and streamlined response processes.

ACADEMIC PROJECTS

Malware Reverse Engineering

- Using static analysis with Hex Editor, **IDA Pro**, and **YARA** rules, examined PE headers, disassembled binaries, and IOC strings, which uncovered **15+ critical vulnerabilities**, stealth persistence mechanisms, and malicious API calls.
- Employing dynamic analysis via Process Monitor, Process Explorer, and **Wireshark**, tracked system behavior to reveal **12 unauthorized registry modifications**, suspicious DLL injection attempts, and anomalous file operations, strengthening forensic understanding of the malware’s evasion tactics.
- Applying reverse engineering workflows with **FakeDNS** and execution flow mapping, intercepted malicious C2 DNS requests, correlated behavioral anomalies, and developed targeted countermeasures, achieving a **60% reduction** in malware infection risk through proactive remediation strategies.

Practical Hard-Drive Forensic Examination – Expert Witness Report

- Using **FTK Imager** with MD5/SHA-1 verification and WiebeTech UltraDock write-blocker, acquired a forensically sound image of a 40 GB Seagate Barracuda HDD, resulting in validated bit-level accuracy and preservation of chain of custody.
- Leveraging **Autopsy 4.22.1** with metadata, file carving, and browser artifact analysis, recovered emails, financial spreadsheets, and jailbreak traces, which tied suspects to iPhone resale operations and revealed **19 flagged CSAM images** and **14 adult-content artifacts**.
- Applying digital forensic methodologies including registry hive analysis, VPN configuration discovery, and user attribution profiling, correlated dual user accounts (Eddie and Jimmy) to distinct roles in fraud, DRM circumvention, and anonymization, directly supporting federal charges under **CFAA**, **Wire Fraud**, and **DMCA** statutes.

ACHIEVEMENTS

C2C CTF 2025 Finalist – International Capture The Flag (CTF) Competition

- Finalist in the global C2C CTF organized by **INCS-CoE** and hosted by **Northeastern University**, solving challenges in cryptography, OSINT, web exploitation, reverse engineering, and forensics while demonstrating expertise in adversarial tactics, red teaming, and offensive security assessments through collaborative multi-domain engagements.

Research Intern – Research Centre Imarat (RCI), DRDO

- Conducted simulation research on Brushless DC motor control systems using MATLAB-Simulink, integrating concepts of cyber-physical systems security, threat modeling for control logic, and anomaly detection in motor telemetry, which enhanced resilience strategies for critical infrastructure and defense-grade embedded systems.

CERTIFICATIONS

- Google Cybersecurity - Coursera
- Junior Penetration Tester (eJPT) - INE Security.
- Certified Cloud Associate (ICCA) - INE Security
- Offensive Hacking Unfolded - The Beginner’s Edition - comproavi.
- Cyber Threat Intelligence 101 - arcX